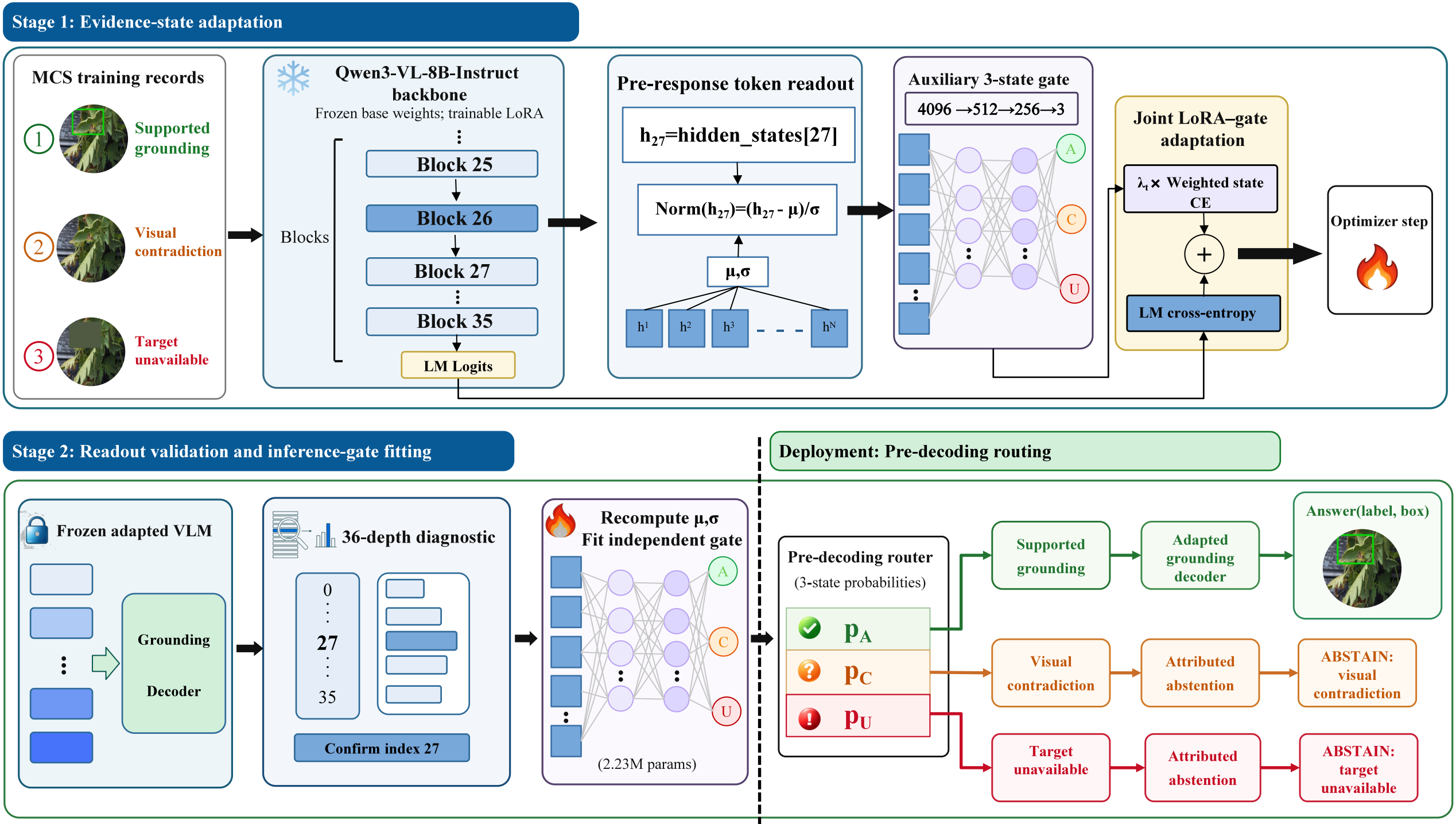

Stage 1: Evidence-state adaptation
MCS training records
1
Supported grounding
2
Visual contradiction
3
Target unavailable
Qwen3-VL-8B-Instruct backbone
Frozen base weights; trainable LoRA
Blocks
Block 25
Block 26
Block 27
Block 35
LM Logits
Pre-response token readout
h27=hidden_states[27]
Norm(h27)=(h27 - μ)/σ
μ,σ
h1
h2
h3
hN
Auxiliary 3-state gate
4096 →512→256→3
A
C
U
Joint LoRA–gate adaptation
λt × Weighted state CE
LM cross-entropy
Optimizer step
Stage 2: Readout validation and inference-gate fitting
Frozen adapted VLM
Grounding Decoder
36-depth diagnostic
0
27
35
Confirm index 27
Recompute μ,σ
Fit independent gate
(2.23M params)
Deployment: Pre-decoding routing
Pre-decoding router
(3-state probabilities)
pA
pC
pU
Supported grounding
Adapted grounding decoder
Answer(label, box)
Visual contradiction
Attributed abstention
ABSTAIN: visual contradiction
Target unavailable
Attributed abstention
ABSTAIN: target unavailable